Adaptive Closed-Loop Hydrogen Recycling System for NdFeB Permanent Magnets from E-Waste

Proposed Classification: Class 15 — Machines and mechanical processing systems

Figure 1: Functional System Architecture

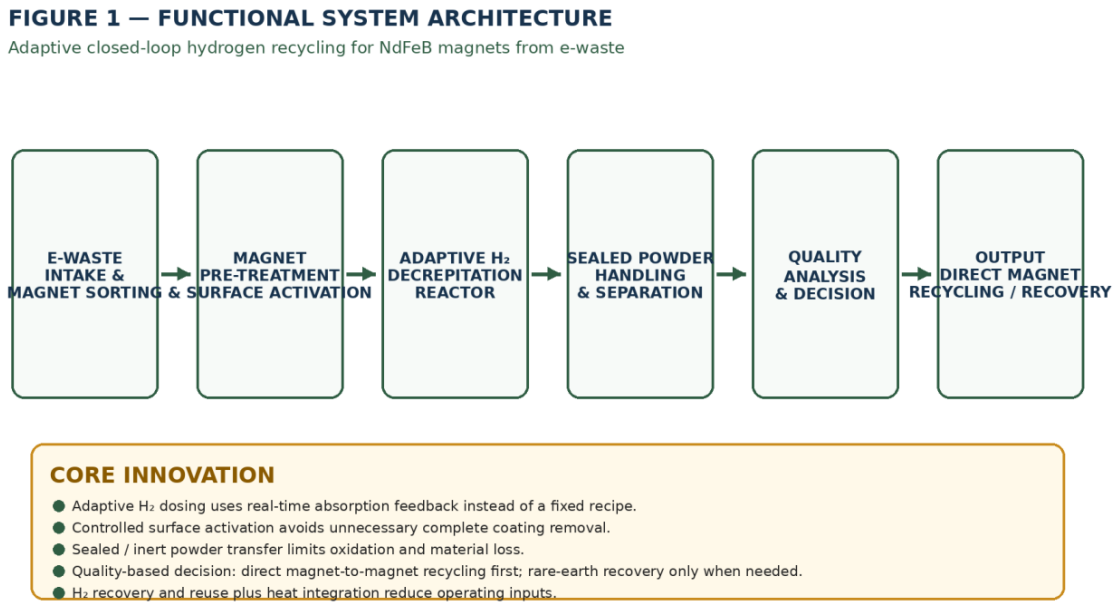


Figure 1 illustrates the functional architecture of the proposed adaptive closed-loop hydrogen recycling system. The system is designed to recover NdFeB permanent magnets from e-waste and convert suitable recovered material into reusable magnet feedstock while minimizing hydrogen consumption, oxidation, energy use and material loss. The process combines e-waste intake and magnet sorting, controlled magnet pre-treatment, an adaptive hydrogen decrepitation reactor, sealed powder handling, quality analysis and a decision pathway for either direct magnet-to-magnet recycling or selective rare-earth recovery.

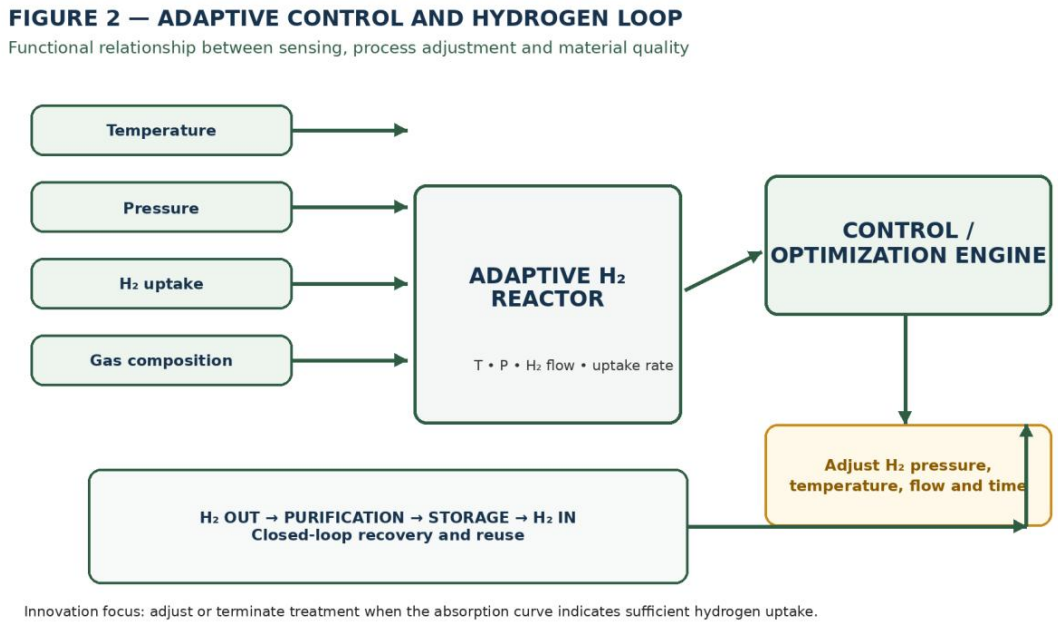
The principal innovation is the integration of real-time hydrogen-uptake feedback with closed-loop hydrogen recovery and sealed powder handling. Instead of operating the hydrogen reactor using a fixed treatment recipe, the proposed system adjusts hydrogen pressure, temperature, flow and treatment duration according to process measurements. This adaptive operation is intended to reduce unnecessary hydrogen use and processing time while maintaining the required powder quality.

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Figure 2: Adaptive Control and Hydrogen Recovery Loop



As shown in Figure 2, the proposed system introduces a feedback-controlled hydrogen treatment cycle. Sensors monitor temperature, pressure, hydrogen flow, gas composition and the hydrogen uptake behavior of the magnet batch. A control and optimization unit uses these measurements to adjust reactor operating conditions and to terminate or modify treatment when sufficient hydrogen uptake has been achieved.

A second innovation is the closed-loop hydrogen pathway. Hydrogen leaving the reactor is directed to purification and storage before being returned to the reactor for subsequent processing. The powder is transferred through a sealed or inert handling path to reduce oxidation after decrepitation. The system may additionally integrate waste heat from thermal processing stages for preheating, thereby reducing external energy demand. These features are intended to improve operating efficiency and process consistency.

The combination of adaptive process control, hydrogen recovery and protected powder handling forms an integrated control architecture rather than treating hydrogen decrepitation as an isolated batch operation.

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Figure 3: Quality-Based Material Decision Pathway

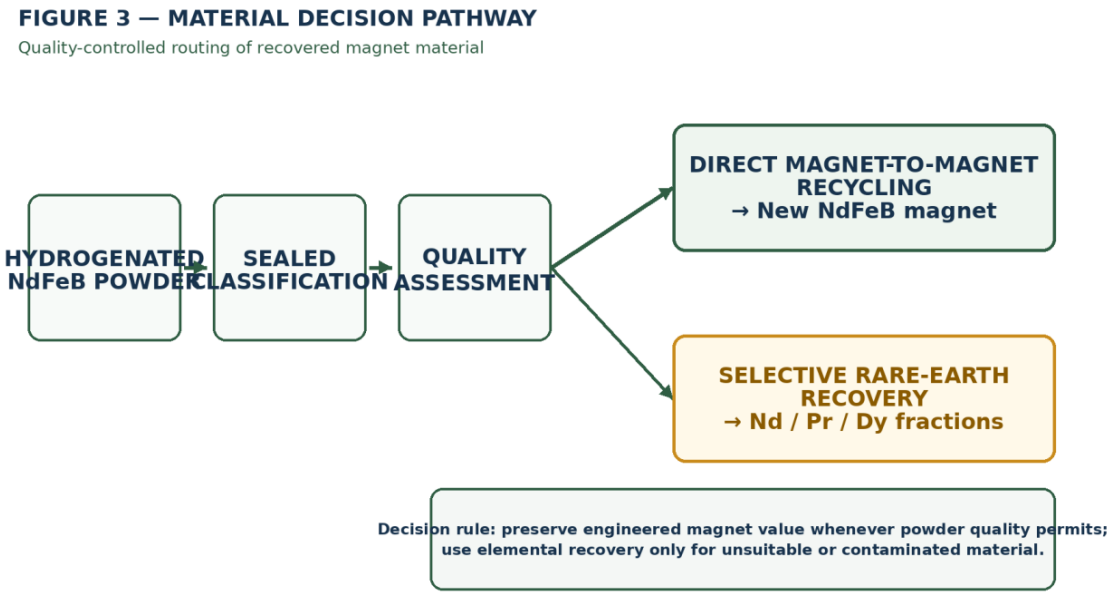


Figure 3 illustrates the quality-based decision pathway of the invention. After hydrogen decrepitation, the recovered NdFeB powder is handled in a controlled environment and classified before quality assessment. The quality assessment may consider composition, oxygen exposure, particle characteristics and other process-defined indicators relevant to producing a recycled magnet.

The proposed decision rule gives priority to direct magnet-to-magnet recycling when the recovered powder is suitable for reuse. Material that is contaminated, compositionally unsuitable or otherwise unable to meet the required recycled-magnet specification can instead be routed to selective rare-earth recovery. This prevents unnecessary destruction of usable engineered magnet material and allows different e-waste feedstocks to be processed according to their value and quality.

To the best of our knowledge and belief, the proposed innovation lies in the specific integration of adaptive hydrogen dosing based on real-time uptake behavior, controlled surface activation, sealed/inert powder transfer, closed-loop hydrogen recovery, quality-based routing and direct magnet-to-magnet priority within a single modular recycling system. This document is a patent-development concept; formal novelty, inventive-step and prior-art assessment would be required before filing.